Deployment manual for PlantChoice Project

Contents

1 Source Code

The source code is hosted on GitHub:
https://github.com/COMP693-Projects-25S2/COMP693_25S2_project_GreenMean

Note: This is a private repository shared with **lincolnmac** account (https://github.com/lincolnmac). Only this account **lincolnmac** has the access permission. Feel free to check with Elizabeth.

2 Clone Source Code

On the server, create a new folder, e.g. PlantChoice. Run the following command in command line

```
cd PlantChoice
git clone --depth 1 https://github.com/COMP693-Projects-25S2/COMP693_25S2_project_GreenMean.git .
```

Note: The command above clones the latest version without history. This will keep the folder size small. To clone the full history, run the same command without --depth 1.

```
git clone https://github.com/COMP693-Projects-25S2/COMP693_25S2_project_GreenMean.git .
```

3. Install Python libraries

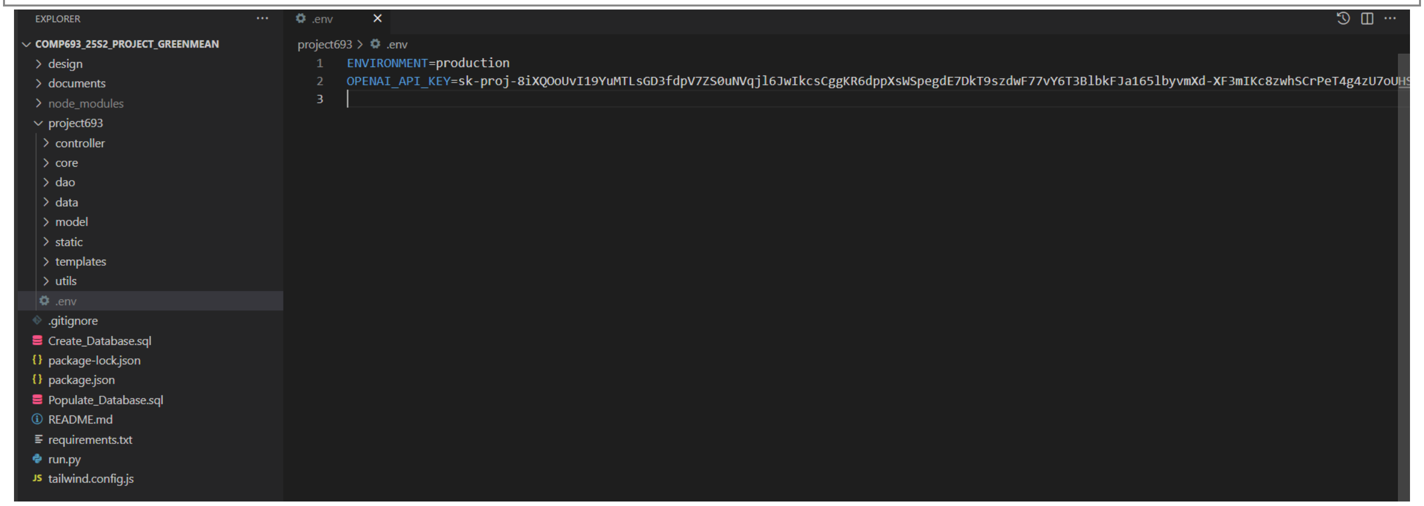
Run the command below at the parent folder, to install Python library dependencies.

```
cd path/to/your/project
pip install -r requirements.txt
```

4. Add .env file

Under the **project693** folder, create a new **.env** file.
The content of the **.env** file is as below:

```
ENVIRONMENT=production
OPENAI_API_KEY=sk-proj-
8iXQOoUvI19YuMTLsGD3fdpV7ZS0uNVqjl6JwIkcsCggKR6dppXsWSpegdE7DkT9szdwF77vY6T3BlbkFJa165lbyvmXd-
XF3mIKc8zwhSCrPeT4g4zU7oUHS8tIgNU_kWZmjIr7l7F1X_5olBfIGXcVamAA
```



The OPENAI_API_KEY is required to fetch AI-generated plant descriptions from OpenAI. If the key expires, the application will not fetch new description but will continue running. Replace with a new key to resume the functionality.

5. Create MySQL database

On the server, create a new database, e.g. named **plantchoice** (or with any another name).
Copy and run SQL commands from the file **Create_Database.sql**, to create tables in database.

6. Insert initialize data

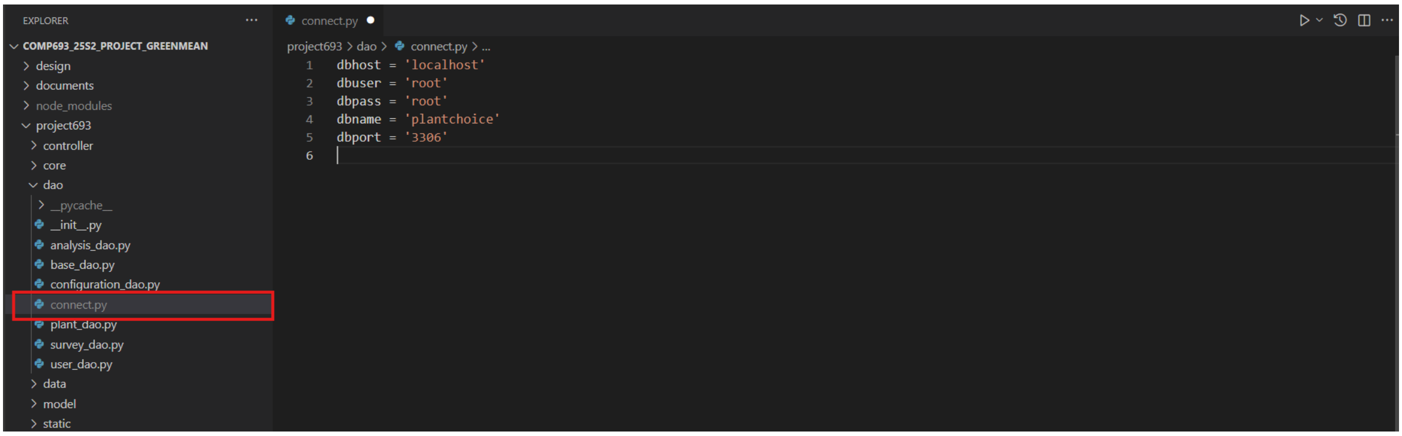
Copy and run SQL commands from the file **Populate_Database.sql**, to insert some initial data, including the image.

The initial image files are under **project693 -> static -> img**. The images files are already copied when cloning from GitHub repository.

7. Add database connection file

Under folder **project693 -> dao**, create a new file **connect.py**
configure database host, username, password, database name and port. Change accordingly based on the server environment.

```python
dbhost = 'localhost'
dbuser = 'root'
dbpass = 'root'
dbname = 'plantchoice'
dbport = '3306'
```



8. Run application

Run the command in the project's root folder

```
python run.py
```

The application should be started and is accessible through http://127.0.0.1:5000

```
c:\root\dev\src\plantchoice (main -> origin)
(venv-693) λ python run.py
 * Serving Flask app 'project693.controller'
 * Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on http://127.0.0.1:5000
Press CTRL+C to quit
 * Restarting with stat
 * Debugger is active!
 * Debugger PIN: 132-834-810
```

Note: The command above runs the project on development server. When deploying to a cloud environment, we usually need to configure the project folder and server settings according to the providers documentation.

Below is an example of how to configure a WSGI file on PythonAnywhere.

```python
# This file contains the WSGI configuration required to serve up your
# web application at http://<your-username>.pythonanywhere.com/
# It works by setting the variable 'application' to a WSGI handler of some
# description.
#
# The below has been auto-generated for your Flask project

import sys

# add your project directory to the sys.path
project_home = '/home/GreenMean/COMP693_25S2_project_GreenMean'
if project_home not in sys.path:
    sys.path = [project_home] + sys.path

# import flask app but need to call it "application" for WSGI to work
from run import app as application  # noqa
```